BreakAPlate Error Log

In a breakAPlate program I had only one error, and it happened because I was rewriting the code from the book which used the gamebooth library which I didn't have. I deleted it and the error disappeared.

```
JFrame frame;
GameBooth breakAPlate;

/**
 * Launch the application.
 */
public static void main(String[] args) {
    EventQueue.invokeLater(new Runnable() {
        public void run() {
            try {
                BrakeAPlate window = new BrakeAPlate();
                window.frame.setVisible(true);
            } catch (Exception e) {
                e.printStackTrace();
            }
        }
    });
}

/**
 * Create the application.
 */
public BrakeAPlate() {
    initialize();
    breakAPlate = new GameBooth(0, FIRST_PRIZE, CONSOLATION_PRIZE);
}
```

TicTacToe Error Log

TicTacToe was a hard program for me as I decided to write it from scratch without copying it from the textbook. My first error appeared when I put the turn variable in the initialize() method, so it was assigned to 0 every time the action happened. I fixed it by assigning the variable in the TicTacToe class.

```java
    private void initialize() {

        //turn variable
        private int turn = 0;

        frame = new JFrame();
        frame.getContentPane().setBackground(new Color(0, 51, 51));
        frame.setBounds(100, 100, 440, 342);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.getContentPane().setLayout(null);

        JLabel winner = new JLabel("");
        winner.setFont(new Font("Snap ITC", Font.PLAIN, 14));
        winner.setForeground(Color.BLACK);
        winner.setHorizontalAlignment(SwingConstants.CENTER);
        winner.setBounds(10, 259, 403, 33);
        frame.getContentPane().add(winner);



            JButton btn1 = new JButton(" ");
            btn1.addActionListener(new ActionListener() {
                public void actionPerformed(ActionEvent e)
                {

                    if(turn == 0 && !btn1.equals("O") )
                    {
                        btn1.setText("X");
                        winner.setText("Second player turn(O)");
                        turn += 1;
                    }
                    /*
                    else if(turn == 1 && !btn1.equals("X"))
                    {
                        btn1.setText("O");
                        turn -= 1;
                    }
```

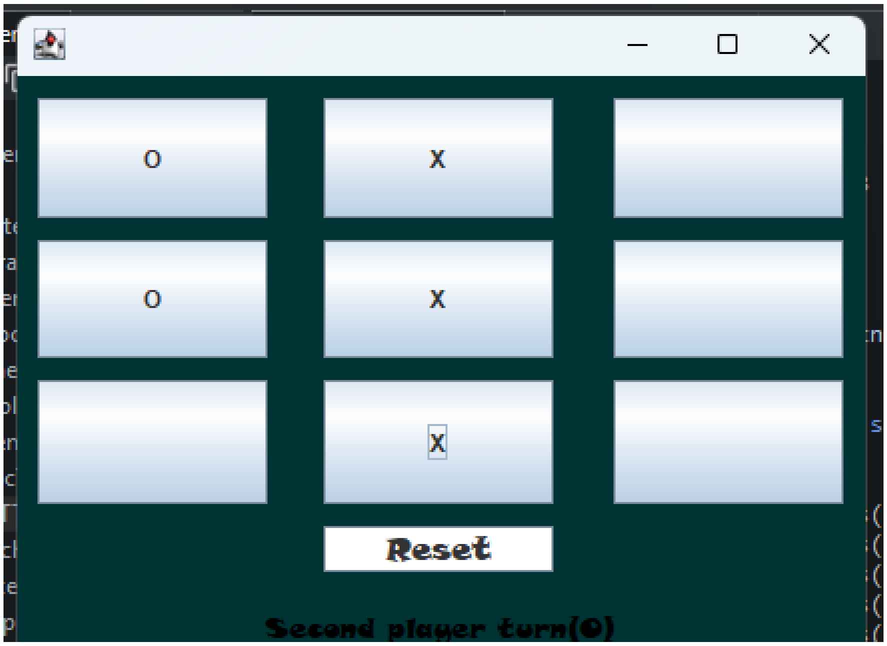
```java
public class TicTacToe {

    // These strings represent th
    public String b1 = " ",b2 = "
    // turn = 0 means it's X's tu
    private int turn = 0;
```

My second error occurred when I decided to write another class TTT to check for the winning combination, but it didn't work as I had to pass all variables with the text on the button(b1, b2, b3 ….). To fix this problem I deleted class TTT entirely and wrote a method inside the TicTacToe class.

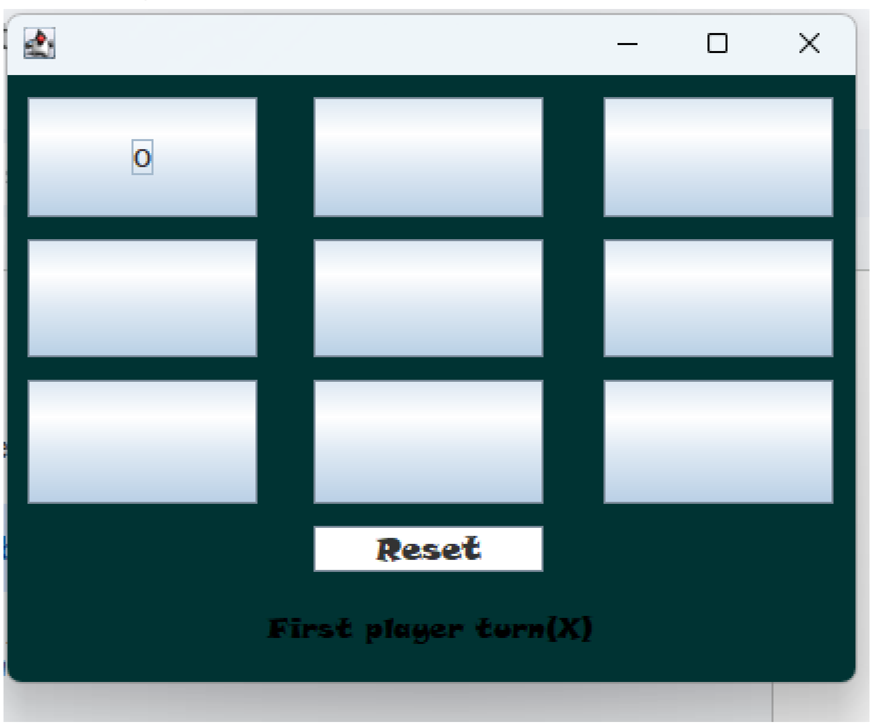
TicTacToe.java LocalBank.java *TTT.java

```java
package chapter10;

public class TTT {

    public String checkWinner(boolean start, String winner)
    {

        if(b1.equals("X") && b2.equals("X") && b3.equals("X")) return "The Winner Is X!";
        if(b4.equals("X") && b5.equals("X") && b6.equals("X")) return "The Winner Is X!";
        if(b7.equals("X") && b8.equals("X") && b9.equals("X")) return "The Winner Is X!";
        if(b1.equals("X") && b5.equals("X") && b9.equals("X")) return "The Winner Is X!";
        if(b7.equals("X") && b5.equals("X") && b3.equals("X")) return "The Winner Is X!";
        if(b1.equals("X") && b4.equals("X") && b7.equals("X")) return "The Winner Is X!";
        if(b2.equals("X") && b5.equals("X") && b8.equals("X")) return "The Winner Is X!";
        if(b3.equals("X") && b6.equals("X") && b9.equals("X")) return "The Winner Is X!";

        if(b1.equals("O") && b2.equals("O") && b3.equals("O")) return "The Winner Is O!";
        if(b4.equals("O") && b5.equals("O") && b6.equals("O")) return "The Winner Is O!";
        if(b7.equals("O") && b8.equals("O") && b9.equals("O")) return "The Winner Is O!";
        if(b1.equals("O") && b5.equals("O") && b9.equals("O")) return "The Winner Is O!";
        if(b7.equals("O") && b5.equals("O") && b3.equals("O")) return "The Winner Is O!";
        if(b1.equals("O") && b4.equals("O") && b7.equals("O")) return "The Winner Is O!";
        if(b2.equals("O") && b5.equals("O") && b8.equals("O")) return "The Winner Is O!";
        if(b3.equals("O") && b6.equals("O") && b9.equals("O")) return "The Winner Is O!";

        return " ";

    }
}
```

```java
    public String checkWinner(boolean start, String winner)
    {

        if(b1.equals("X") && b2.equals("X") && b3.equals("X")) return "The Winner Is X!";
        if(b4.equals("X") && b5.equals("X") && b6.equals("X")) return "The Winner Is X!";
        if(b7.equals("X") && b8.equals("X") && b9.equals("X")) return "The Winner Is X!";
        if(b1.equals("X") && b5.equals("X") && b9.equals("X")) return "The Winner Is X!";
        if(b7.equals("X") && b5.equals("X") && b3.equals("X")) return "The Winner Is X!";
        if(b1.equals("X") && b4.equals("X") && b7.equals("X")) return "The Winner Is X!";
        //if(b2.equals("X") && b5.equals("X") && b8.equals("X")) return "The Winner Is X!";
        //if(b3.equals("X") && b6.equals("X") && b9.equals("X")) return "The Winner Is X!";

        if(b1.equals("O") && b2.equals("O") && b3.equals("O")) return "The Winner Is O!";
        if(b4.equals("O") && b5.equals("O") && b6.equals("O")) return "The Winner Is O!";
        if(b7.equals("O") && b8.equals("O") && b9.equals("O")) return "The Winner Is O!";
        if(b1.equals("O") && b5.equals("O") && b9.equals("O")) return "The Winner Is O!";
        if(b7.equals("O") && b5.equals("O") && b3.equals("O")) return "The Winner Is O!";
        if(b1.equals("O") && b4.equals("O") && b7.equals("O")) return "The Winner Is O!";
        //if(b2.equals("O") && b5.equals("O") && b8.equals("O")) return "The Winner Is O!";
        //if(b3.equals("O") && b6.equals("O") && b9.equals("O")) return "The Winner Is O!";

        return " ";

    }
        private void disableButtons(){
```



During this process, I also forgot to include 4 combinations which resulted in error when running the program, so to fix it I just added the needed combination(shown as comments in previous screenshot).



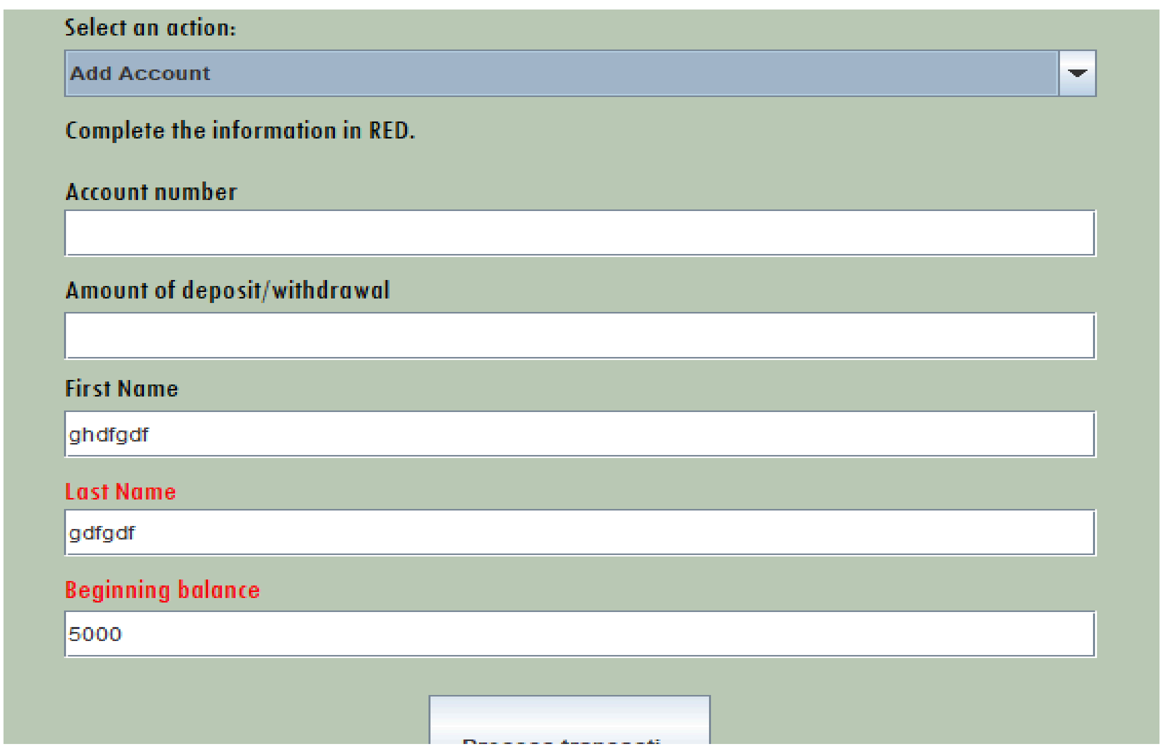
The last error occurred when the first game had been completed, the Play Again button didn't change the turn to X. To fix it I just assigned the turn variable to 0 (X) if the button is clicked.

```
else if(Start.equals("Reset"))
{
    btn1.setVisible(false);
    btn2.setVisible(false);
    btn3.setVisible(false);
    btn4.setVisible(false);
    btn5.setVisible(false);
    btn6.setVisible(false);
    btn7.setVisible(false);
    btn8.setVisible(false);
    btn9.setVisible(false);
    //turn = 0;
    b1 = " ";
    b2 = " ";
    b3 = " ";
    b4 = " ";
    b5 = " ";
    b6 = " ";
    b7 = " ";
    b8 = " ";
    b9 = " ";
    btn1.setText(" ");
    btn2.setText(" ");
    btn3.setText(" ");
    btn4.setText(" ");
    btn5.setText(" ");
    btn6.setText(" ");
    btn7.setText(" ");
    btn8.setText(" ");
    btn9.setText(" ");
     start = false;
    startbtn.setText("Start");
    winner.setText(" ");
}
```

LocalBank Error Log

I found the first error when I was trying to get the user's input when the user chose an action "Add Account" , and only the first name label was changing the color to black. To fix it I rewrote the code to get the input after the button is pressed and not when the action "Add Account" is chosen.

```
JComboBox action = new JComboBox();
action.setModel(new DefaultComboBoxModel(new String[] {"", "Add Account", "Delete Account", "Deposit", "Withdrawal", "Check Balance"}));
action.addItemListener(new ItemListener() {
    public void itemStateChanged(ItemEvent e)
    {
        if(action.getSelectedItem().equals("Add Account"))
        {

            if(!(firstN.getText().equals("")))
            {
                fname = firstN.getSelectedText();
                firstNlbl.setForeground(new Color(0,0 ,0));
            }
            else if(!(lastN.getText().equals("")))
            {
                lastNlbl.setForeground(new Color(0,0 ,0));
                lname = lastN.getSelectedText();
            }
            else if (!(balance.getSelectionStart() >= 0))
            {
                bal = balance.getSelectionStart();
                balancelbl.setForeground(new Color(0,0 ,0));
            }
            if(!(firstN.getText().equals("")) && !(lastN.getText().equals("")) && !(balance.getSelectionStart() >= 0))
            {
                bank.addAccount(fname, lname, bal);
            }
            else
            {
                firstNlbl.setForeground(new Color(255,0 ,0));
                lastNlbl.setForeground(new Color(255,0 ,0));
                balancelbl.setForeground(new Color(255,0 ,0));

            }
```

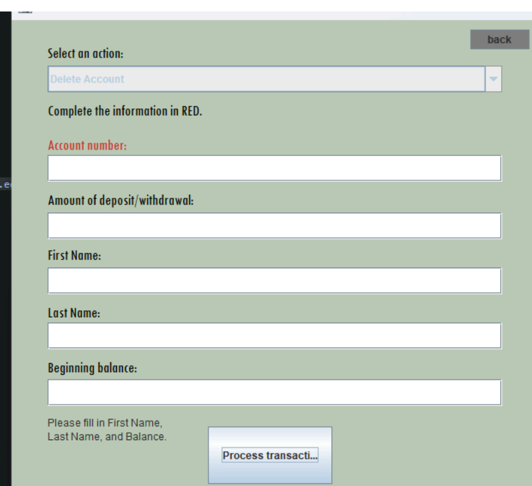


“Fixed code”.

```java
process = new JButton("Process transaction");
process.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e)
    {
        // Read and trim all input values from the text fields

        String fNameInput = firstN.getText().trim();
        String lNameInput = lastN.getText().trim();
        String balInput = balance.getText().trim();
        String accountInput = account.getText().trim();
        String amountInput = amount.getText().trim();

        // Validate required fields for adding an account

        if((fNameInput.isEmpty() || lNameInput.isEmpty() || balInput.isEmpty()) && action.getSelectedItem().equals("Add Account"))
        {
            accountinfo.setText("Please fill in First Name, \nLast Name, and Balance.");
            return;
        }
        try {

            // If the user selected "Add Account", create a new account

            if(action.getSelectedItem().equals("Add Account"))
            {
                // Convert balance string to double (may throw NumberFormatException)

                double balValue = Double.parseDouble(balInput);

                // Create the account using your bank object and get its ID

                String id = bank.addAccount(fNameInput, lNameInput, balValue);

                // Show success message with the new account ID

                accountinfo.setText("Success! \nAccount ID: " + id +
                "\nKeep this ID safe.");

                // Reset label colors back to black

                firstNlbl.setForeground(Color.BLACK);
                lastNlbl.setForeground(Color.BLACK);
                balancelbl.setForeground(Color.BLACK);

                // Clear fields and reset combo box

                cleanTextFields();
                action.setEnabled(true);
                action.setSelectedItem("");

                // Disable all text fields again

                disableTextFields();
```

The second error occurred when the code that is assigned to the action “Add account” was running when other actions were chosen(look at the bottom left corner of the program). To fix this I just had to add brackets in the if statement. The if statement was checking if the text fields were empty and the selected action was “Action”, but it runned if just when Firstname or Lastname text field was empty.

```java
        }
    });
    action.setBounds(42, 53, 530, 31);
    panel.add(action);

    JButton process = new JButton("Process transaction");
    process.addActionListener(new ActionListener() {
        public void actionPerformed(ActionEvent e)
        {
            String fNameInput = firstN.getText().trim();
            String lNameInput = lastN.getText().trim();
            String balInput = balance.getText().trim();
            String accountInput = account.getText().trim();
            if(fNameInput.isEmpty() || lNameInput.isEmpty() || balInput.isEmpty() && action.getSelectedItem().e
                accountinfo.setText("Please fill in First Name, \nLast Name, and Balance.");
                return;
            }
            try {
                double balValue = Double.parseDouble(balInput);

                // 3. Create the account using your bank object
                String id = bank.addAccount(fNameInput, lNameInput, balValue);

                accountinfo.setText("Success! \nAccount ID: " + id +
                                    "\nKeep this ID safe.");

                // Reset colors
                firstNlbl.setForeground(Color.BLACK);
                lastNlbl.setForeground(Color.BLACK);
                balancelbl.setForeground(Color.BLACK);

            } catch (NumberFormatException ex) {
                accountinfo.setText("Error: Balance must be \na valid number.");
            }
            if(accountInput.isEmpty() && action.getSelectedItem().equals("Delete Account")) {
                accountinfo.setText("Please fill in Account ID.");
                return;
            }
            else
            {
                accountinfo.setText("" + bank.deleteAccount(accountInput));
            }
```




```java
    if(   (   fNameInput.isEmpty() || lNameInput.isEmpty() || balInput.isEmpty()   ) && action.getSelectedItem().equals("Add Account")) {
```